

Question Number	Error	Line Number	Error Type	Error Correction
1	<code>if(totalAccounts > 10)</code>	88	Logical Error (Incorrect Bounds Check)	<code>accounts[]</code> has size 10 (valid indices 0-9), but the check allows <code>totalAccounts</code> to reach 10 before flagging "Database Full," causing an out-of-bounds write. Check before writing and use: <code>if(totalAccounts >= 10) { printf("Database Full\n"); return; }</code>
2	<code>for(i=0;i<=totalAccounts;i++)</code>	98	Logical Error (Off-by-One)	Using <code><=</code> accesses one element past the last valid account. Use: <code>for(i=0;i<totalAccounts;i++)</code>
3	<code>accounts[acc].balance = accounts[acc].balance + amount;</code>	123	Runtime Error (Unvalidated Array Index)	<code>acc</code> comes directly from user input and is never checked against valid bounds before use. Validate first: <code>if(acc >= 0 && acc < totalAccounts)</code> <code>accounts[acc].balance += amount;</code>
4	<code>if(accounts[acc].balance > amount) { printf("Insufficient Balance\n"); } else { withdraw }</code>	139	Logical Error (Inverted Condition)	The condition is backwards: it prints "Insufficient Balance" when funds ARE sufficient, and withdraws when they are NOT. Swap the branches: <code>if(accounts[acc].balance < amount) printf("Insufficient Balance\n"); else { accounts[acc].balance -= amount; }</code>
5	<code>totalAccount</code>	192	Semantic Error (Undeclared Identifier)	Type: the actual variable is "totalAccounts". Correct it to: <code>totalAccounts</code>
6	<code>calculateInterest();</code>	194	Semantic Error (Undeclared/Undefined Function)	No function named <code>calculateInterest()</code> is declared or defined; the actual function in the file is named <code>interestCalculator()</code> . Call: <code>interestCalculator();</code> or rename the function.
7	<code>a.accNo = "1001";</code>	203	Type Mismatch Error	<code>accNo</code> is an int; cannot assign a string literal. Use: <code>a.accNo = 1001;</code>
8	<code>a.balance = "50000";</code>	207	Type Mismatch Error	<code>balance</code> is a float; cannot assign a string literal. Use: <code>a.balance = 50000;</code>
9	<code>accounts[accNo].balance = accounts[accNo].balance + 1000;</code>	218	Runtime Error (Unvalidated Array Index)	<code>accNo</code> comes directly from user input and is never checked against valid bounds before use. Validate first: <code>if(accNo >= 0 && accNo < totalAccounts)</code> <code>accounts[accNo].balance += 1000;</code>
10	<code>accounts[accNo] = NULL;</code>	228	Compile Error (Type Mismatch)	<code>accounts[accNo]</code> is a struct, not a pointer; <code>NULL</code> cannot be assigned to it. Also <code>accNo</code> is never bounds-checked. Reset the struct instead, e.g. <code>memset(&accounts[accNo], 0, sizeof(struct Account));</code> after validating <code>accNo</code> .
11	<code>if(name == accounts[0].name)</code>	237	Logical Error	This compares array addresses, not the string contents. Use: <code>if(strcmp(name, accounts[0].name) == 0)</code>
12	<code>average = generateAverage();</code>	247	Semantic Error (Undeclared/Undefined Function)	<code>generateAverage()</code> is never declared or defined anywhere in the file. Define it, or remove/replace the call.
13	<code>fp = fopen("accounts.txt","r");</code>	256	Logical Error (Wrong File Mode)	The file is opened in read-only mode ("r") but then written to with <code>fprintf</code> . Open it for writing instead: <code>fp = fopen("accounts.txt","w");</code> and check <code>fp != NULL</code> before using it.
14	<code>fp = fopen("missing.txt","r");</code>	267	Runtime Error (Missing NULL Check)	If the file does not exist, <code>fopen</code> returns <code>NULL</code> , and the following <code>fscanf(fp,...)</code> call will crash. Add a check: <code>if(fp == NULL) { printf("File not found\n"); return; }</code>
15	<code>arr[10] = 500;</code>	311	Runtime Error (Array Index Out of Bounds)	<code>arr</code> has size 5 (valid indices 0-4). Use a valid index, e.g. <code>arr[4] = 500;</code>
16	<code>*ptr = 100;</code>	315	Runtime Error (Null Pointer Dereference)	<code>ptr</code> is <code>NULL</code> ; dereferencing it crashes the program. Allocate memory first, e.g. <code>ptr = malloc(sizeof(int));</code>
17	<code>printf("%d\n",a/b);</code>	320	Runtime Error (Division by Zero)	<code>b</code> is 0, causing undefined behavior/crash. Check the divisor before dividing: <code>if(b != 0) printf("%d\n", a/b);</code>